HW1 Cover Page

MAE 3240 Spring 26

Name: Emily Gamez-Garcia Date of Submission: 2/3/26
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Acknowledgement: TAs from Mon & Tues OH, Desmos

HW1 P1

Problem 1

When the Spacecraft is in Space outside the atmosphere, there is radiation from stars and planets being emitted onto the Spacecraft and conduction between the inside and outside layers of the spacecraft ☆

When the Spacecraft is moving across the atmosphere, there is still radiation and conduction, and now there is also convection because the atmosphere allows for air to circulate ★

HW1 P2

Problem 2

Knowns:

$k = 150 \frac{W}{mK}$

$w = 5mm \rightarrow 0.005m$

$t = 1mm \rightarrow 0.001m$

$q = 9W$

$T_\infty = 20°C \rightarrow 293.15°K$

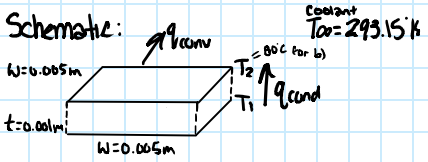
Just for b) $T_2 = 80°C \rightarrow 353.15°K$

Find:

a) Steady State temp diff of top & bottom Surfaces

b) minimum h when $T_2 < 80°C$ and convection process

Schematic:



ASSUMTIONS:

- Steady State
- Side + back Surfaces thermally insulated
- thermal radiation negligible

Analysis:

a) $\dot{E}_{in} + \cancel{\dot{E}_g} - \dot{E}_{out} = \cancel{\dot{E}_{st}}$

$\dot{E}_{in} - \dot{E}_{out} = 0$ } bc steady state

$\dot{E}_{in} = \dot{E}_{out}$

$q_{cond} = q_{conv}$, $q = 9W$

$-kA\frac{T_2 - T_1}{t} = hA(T_2 - T_\infty)$

$-9W = 150 \frac{W}{mK} \cdot (0.005m)^2 \cdot \frac{T_2 - T_1}{0.001m}$

$-9W = 3.75 \frac{W}{K}(T_2 - T_1)$

$T_2 - T_1 = -2.4°K$ ★

b) Recall: $q_{cond} = q_{conv} = 9W$

$9W = q_{conv}$

$9W = hA(T_2 - T_\infty)$

$h = \frac{9W}{A(T_2 - T_\infty)}$

$h = \frac{9W}{(0.005m)^2(353.15°K - 293.15°K)}$

$h = 6000 \frac{W}{m^2K}$, forced convection because h is not in the free convection liquid range ★

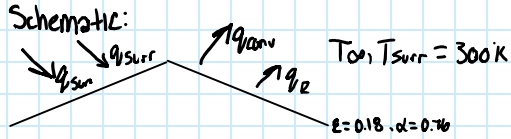
HW1 P3

Problem 3

Knowns:
$\varepsilon = 0.18$
$\alpha = 0.76$
$G = 750 \frac{W}{m^2}$
$T_\infty = 300\,K$
$T_{surr} = 300\,K$
$h = 10 \frac{W}{m^2 K}$

Find:
a) Temp of roof, safe for cat to walk?
b) dominant heat transfer mode

Schematic:



Assumptions:
- bottom surface of roof is thermally insulated
- roof is gray body to irradiation & surroundings
- dominant heat transfer in b) refers to that coming out of roof

Analysis

a) Out In

$q_{conv} + q_R = q_{surr} + q_{sun}$

$hA(T_R - T_\infty) + \varepsilon A \sigma T_R^4 = \varepsilon A \sigma T_{surr}^4 + \alpha A G$

$10 \frac{W}{m^2 K}(T_R - 300\,K) + 0.18 \cdot 5.67\times10^{-8} \frac{W}{m^2 K^4} \cdot T_R^4 = 0.18 \cdot 5.67\times10^{-8} \frac{W}{m^2 K} \cdot 300\,K^4 + 0.76 \cdot 750 \frac{W}{m^2}$

Plugged & graphed into Desmos to solve for T_R

$T_R = 349.95871\,K$, too hot for cat, not comfortable :(*

b) is q_{conv} or q_R greater?

$h(T_R - T_\infty)$ vs. $\varepsilon \sigma T_R^4$

$10 \frac{W}{m^2 K}(350\,K - 300\,K)$ vs. $0.18 \cdot 5.67\times10^{-8} \frac{W}{m^2 K^4} \cdot 350\,K$

$500 \frac{W}{m^2}$ vs. $153.15 \frac{W}{m^2}$

Convection is the dominant heat transfer mode *